# A Differentiable CVaR-Constrained Allocator and Two Failure Modes in Evaluating Constrained Reinforcement Learning for Finance

Anonymous Author(s)

## Abstract

We present a differentiable Conditional Value-at-Risk (CVaR) constrained portfolio allocator and use it to expose two failure modes that affect any constrained financial RL study. On a seven-ETF macro universe (2008–2024, weekly, 5 bps costs) the constraint acts as an effective tail safety layer. Through the 2020 crash and 2022 selloff it cuts $\text{CVaR}_{99}$ by 49% and max drawdown by 43% while improving Sharpe by 40% and eliminating hard constraint violations, and a 40-fold two-universe walk-forward confirms constrained $\text{CVaR}_{99}$ is lower in 26/40 folds ($p = 8.8\times10^{-6}$). Building the allocator surfaces two transferable methodological lessons. (i) A **Lagrangian constraint-coupling failure**, where standardising the reward advantage but not the cost advantage renders the constraint silently inert for any reasonable multiplier, which we name and fix. (ii) An **evaluation-protocol reversal**, where, on a 31-asset universe, a single stress window ranks the learner above every optimiser while a rolling walk-forward inverts the ranking, so apparent "beat-the-optimiser" claims can be protocol artifacts. We are explicit about scope. The pure learned allocator does not beat rolling minimum-variance in aggregate, though a regime-switching hybrid does (+0.107 Sharpe)—an edge robust across the volatility-quantile axis but lost at a deeper −6% selloff-threshold definition. The contribution is the constraint mechanism and the two evaluation failure modes, not a horse-race win.

## CCS Concepts

• **Computing methodologies** → **Reinforcement learning**; • **Applied computing** → *Economics*.

## Keywords

portfolio allocation, reinforcement learning, Conditional Value-at-Risk, drawdown control, constrained MDP, differentiable optimisation

**ACM Reference Format:**

Anonymous Author(s). 2026. A Differentiable CVaR-Constrained Allocator and Two Failure Modes in Evaluating Constrained Reinforcement Learning for Finance. In *Proceedings of ACM International Conference on AI in Finance (ICAIF '26).* ACM, New York, NY, USA, 8 pages. https://doi.org/10.1145/nnnnnnn.nnnnnnn



*ICAIF '26, Milan, Italy*



ACM ISBN 978-x-xxxx-xxxx-x/YYYY/MM
https://doi.org/10.1145/nnnnnnn.nnnnnnn

## 1 Introduction

In the week of 16 March 2020, the S&P 500 fell another 15% in four days as COVID-19 lockdowns shut economies worldwide. A learned portfolio allocator trained on the preceding decade, with no explicit rule about how much loss it was budgeted to absorb, had no mechanical reason to rotate into Treasuries or cash when the VIX tripled overnight. Absent a tail limit, the policy kept holding equities through the worst losses. An otherwise-identical allocator carrying an explicit Conditional Value-at-Risk (CVaR) limit had already rotated into the cash/rates leg before the trough and emerged with roughly half the drawdown (11.8% peak-to-trough versus 20.6% for its unconstrained twin, Table 4).

That difference is a result, not the headline. An explicit CVaR constraint is a safety layer that cuts tail losses of a *learned* allocator without sacrificing risk-adjusted return. There is, however, a catch. In the first model-free implementation we tried, a standard Lagrangian actor–critic, the *dual channel was inert*: the Lagrange multiplier stayed pinned near zero, so the constraint exerted no explicit tail pressure through the mechanism that was supposed to enforce it. The culprit is a subtle scale mismatch that we describe, name, and fix in Section 5. It is likely present in any Lagrangian RL system that standardises the return advantage but not the cost advantage. (As Section 5 shows, the constrained *architecture* still regularises the tail through its safety-critic and projection structure even when the multiplier is inert, the policies are *not* numerically identical, but the Lagrangian mechanism itself never engages.)

Reinforcement learning for portfolio allocation is usually evaluated on return or Sharpe ratio, with risk handled implicitly. Practitioners, however, manage *tail* risk explicitly. Markowitz mean–variance optimisation [11] penalises upside and downside symmetrically and is silent on path-wise drawdown. CVaR is a coherent, convex tail measure [13] that RL can optimise directly. We ask a narrow, falsifiable question.

> *Does an explicit CVaR constraint reduce the downside tail losses and constraint breaches of a learned allocator, versus an otherwise identical unconstrained learner, while remaining competitive with standard portfolio optimisers after costs?*

**We make the following contributions.**

1. **A differentiable CVaR-constrained allocator.** On a leak-free, tested portfolio-control environment (transaction costs, an admissible-set projection, and a rolling CVaR state and constraint), we back-propagate a return objective plus a differentiable CVaR penalty through the rollout, with a Lagrangian dual on a breach-rate budget and a residual policy over an adaptive anchor (Section 4).
2. **A constraint that demonstrably binds, with a defined domain of superiority.** The constraint cuts tail risk without

sacrificing risk-adjusted return, and a regime-switching hybrid (constrained RL in high-volatility weeks, min-variance otherwise) reaches walk-forward Sharpe 1.316 vs 1.209 for pure min-variance, a 2.5× Sharpe advantage in the high-volatility regime where the tail constraint is most active (Sections 7, 8).

(3) **A constraint-coupling failure mode.** In the model-free Lagrangian actor–critic, standardising the reward advantage but not the cost advantage makes the constraint inert for any reasonable multiplier. We name and fix it, a diagnostic that transfers to any constrained financial RL (Section 5).

(4) **An evaluation-protocol reversal.** On a 31-asset universe a single stress window ranks the learned allocator above every optimiser, while a rolling walk-forward inverts the ranking (min-variance Sharpe 1.40 vs 0.74), so apparent "beat-the-optimiser" results can be protocol artifacts (Section 7.9). Our evaluation, deterministic optimisers plus three tuned model-free RL families under a two-universe rolling walk-forward with Deflated-Sharpe and multiple-testing control, is built to avoid exactly this trap.

We are explicit about scope. The pure learned allocator does **not** beat a rolling minimum-variance optimiser in aggregate. The hybrid **does** (+0.107 Sharpe), exploiting the regime in which the tail constraint is most active. But a horse-race win is *not* the contribution. The contribution is methodological and transferable. A differentiable constraint mechanism that demonstrably binds and cuts tails, and two failure modes, constraint-coupling and protocol-reversal, that any constrained-RL-for-finance study must control for or risk reporting an artifact. The allocator is the testbed that makes both failure modes visible and fixable.

## 2 Background and Related Work

*Mean–variance optimisation and its inputs.* MVO [11] is sensitive to input estimates. Decision-focused learning trains return predictors to improve downstream MVO decisions rather than raw forecast accuracy [9], and mean–variance-efficient collaborative filtering blends user preference with portfolio theory [4]. These works improve the *inputs* or *recommendation* layer of a variance-based objective. We instead constrain a *tail* risk measure on a learned policy.

*Conditional Value-at-Risk.* $\mathrm{CVaR}_\alpha$ is the *expected portfolio loss on the worst* $(1-\alpha)$ *fraction of outcomes*, the average of all losses that exceed the $\alpha$-quantile, known as Value-at-Risk ($\mathrm{VaR}_\alpha$). Unlike variance it ignores upside. Unlike VaR it averages across the entire tail rather than reporting a single threshold. Rockafellar and Uryasev [13] show it has the tractable form

$$\mathrm{CVaR}_\alpha(L) \;=\; \min_{\zeta\in\mathbb{R}}\left\{\zeta + \frac{1}{1-\alpha}\,\mathbb{E}[\max(L-\zeta,\,0)]\right\}, \qquad (1)$$

where $L$ denotes the portfolio loss and the inner expectation is over the empirical loss distribution. The minimum is achieved at $\zeta = \mathrm{VaR}_\alpha$, the representation admits a convex LP (we include this as a baseline) and a differentiable penalty via `topk`.

*RL for portfolio management.* The dominant pattern in RL-for-portfolio literature is to train on a single historical window, evaluate on one held-out test split, and compare against equal-weight or buy-and-hold. Jiang et al. [8] introduce the EIIE architecture for cryptocurrency portfolio allocation, comparing against twelve classical strategies on a single ≈50-day test window, no walk-forward, no statistical testing, no tail-risk constraint, and no minimum-variance baseline. Ye et al. [16] augment the state with sentiment signals (SARL) and report higher accumulated return on a fixed evaluation period, again with no cross-validation and no tail-risk metric. Liu et al. [10] (FinRL) provide a reusable RL-for-trading framework that includes a minimum-variance baseline and supports rolling retraining, but the primary demonstrations use a single train/test split and report no significance tests or seed-averaged results. Most closely, Huang [7] applies continuous-time RL to asset–liability management and benchmarks against six classical and RL baselines, but targets a quadratic surplus objective rather than an explicit tail constraint. None of these works uses CVaR as a constraint, seeds multiple runs, or applies statistical testing. *To our knowledge, this is the first work to subject RL portfolio allocation to a 40-fold, two-universe rolling walk-forward with Wilcoxon testing and Benjamini–Hochberg multiple-testing control,* directly exposing the protocol-reversal artefact that single-window evaluations conceal.

*Direct reinforcement and pathwise gradients.* Moody and Saffell [12] first trained a recurrent trading policy by back-propagating the *differential Sharpe ratio* directly through the return function, an exact pathwise gradient predating deep learning. Buehler et al. [2] (Deep Hedging) extend this to neural derivative hedging, back-propagating CVaR as a *utility* (a risk-aversion coefficient) through a simulated trajectory. Our differentiable allocator inherits the pathwise-gradient insight from both, but replaces the risk-aversion coefficient with an explicit CVaR *budget constraint.* The practitioner specifies a tolerable loss threshold directly, and a dual variable enforces it. Model-free policy gradients (A2C/PPO [14]) and off-policy actor–critics (SAC [6]) ignore the differentiable structure of the portfolio reward entirely. Our allocator is strictly more sample-efficient for this problem class. Recent work makes such policies auditable. ProtoHedge [5] replaces the black-box hedger with a prototype-similarity policy at negligible performance cost.

*CVaR-constrained RL.* Chow et al. [3] derive the Lagrangian gradient for CVaR-constrained CMDPs. Tessler et al. [15] (RCPO) and Ying et al. [17] (CPPO) implement hard CVaR constraints for robotics control tasks. None of these works identifies the *constraint-coupling failure* we describe. Standardising return-advantage to unit scale without normalising the cost advantage makes the Lagrange multiplier permanently inert, a failure specific to portfolio allocation where returns are $O(10^{-3})$ and breach indicators are $O(1)$, invisible in robotics. We pose allocation as a constrained MDP [1] and show that the well-scaled dual is what makes the constraint actually bind.

## 3 Problem Formulation

*MDP.* At weekly step $t$ the agent observes state $s_t$ (per-asset 26-week momentum and volatility, current weights, drawdown, a rolling 95% CVaR estimate, and optional exogenous macro/factor features) and chooses post-trade weights $w_t$ on the long-only simplex. The environment projects $w_t$ onto the admissible set $\mathcal{W}$

(max weight 0.40, turnover cap 0.50, optional cash floor), charges proportional cost $c\|w_t - w_{t-1}\|_1$ ($c = 5\,\text{bps}$), and realises $r_t = w_t^\top \rho_t - c\|w_t - w_{t-1}\|_1$, where $\rho_t$ are next-week asset returns. Features at $t$ use only $\rho_{<t}$ (no look-ahead, enforced by tests).

*Constrained objective (CMDP).* A *constrained Markov Decision Process* (CMDP) [1] augments the standard MDP with one or more inequality constraints that the policy must satisfy on average. We maximise risk-adjusted return subject to a tail budget. $\max_\pi J(\pi)$ s.t. $\text{CVaR}_\alpha(\text{loss}) \le d$, with $\alpha = 0.95$. The Lagrangian relaxation converts the constraint into a penalty, $\mathcal{L}(\pi, \lambda) = J(\pi) - \lambda\,(\text{CVaR}_\alpha - d)$, and a dual variable $\lambda \ge 0$ is raised whenever the constraint is violated and lowered when slack remains. In equilibrium $\lambda$ acts as a shadow price for tail risk. A higher $\lambda$ means the policy has been pushing against its tail budget.

## 4 Method

*Differentiable allocator.* The actor is an MLP mapping $s_t$ to simplex weights via softmax. Because the environment's reward and a sample-CVaR are differentiable in $w$, we train by back-propagating, over randomised windows of the training series, the objective

$$\mathcal{L} = -\,\text{Sharpe}(\text{net returns}) \;+\; \lambda\,\text{ReLU}(\widehat{\text{CVaR}} - d) \;+\; \kappa\,\text{turnover}, \tag{2}$$

where $\text{Sharpe} = \bar{r}/\sigma_r$ is the annualised mean return divided by its standard deviation (reward per unit of risk), $\widehat{\text{CVaR}}$ is the mean tail loss over the worst $(1-\alpha)$ fraction of window net returns (a differentiable tail mean via `topk`), and $d$ is the pre-specified tail budget. This pathwise gradient is far more sample-efficient than score-function RL for this problem.

*Pathwise gradient and where the constraint acts.* Every term in $\mathcal{L}$ is a differentiable function of the rollout, so the policy is trained by exact back-propagation through the simplex map and the reward, no score-function estimator. With net returns $r_t = w_t^\top \rho_t - c\|w_t - w_{t-1}\|_1$ and tail set $\mathcal{T} = \{t : R_t \text{ among the worst } \lceil(1-\alpha)T\rceil \text{ of the window}\}$,

$$\frac{\partial \mathcal{L}}{\partial \theta} = \sum_t \Big( \underbrace{-\frac{\partial\,\text{Sharpe}}{\partial r_t}}_{\text{all weeks}} \;-\; \underbrace{\frac{\lambda}{|\mathcal{T}|}\,\mathbf{1}[\widehat{\text{CVaR}} > d]\,\mathbf{1}[t \in \mathcal{T}]}_{\text{tail weeks only}} \Big)\,\frac{\partial r_t}{\partial w_t}\,\frac{\partial w_t}{\partial \theta}, \tag{3}$$

where $\partial w_t/\partial\theta$ flows through the residual softmax (below). The `topk` subgradient routes the dual pressure to *exactly* the worst-$(1-\alpha)$ weeks, and only while the constraint is breached. The penalty lifts the deep-loss weeks without distorting the bulk of the path. This locality is what makes the constraint bind cleanly once the coupling is correctly scaled (Section 5).

*Residual policy over an adaptive anchor.* To generalise across regimes rather than overfit one price path, weights are $w_t = \text{softmax}(\log a_t + f_\theta(s_t))$, where $a_t$ is an adaptive inverse-volatility anchor computed from recent returns. At initialisation $f_\theta \approx 0$, so the policy *is* the anchor (a strong, regime-robust prior). Training learns a state-dependent tilt. Validation model selection keeps the best objective subject to the validation CVaR limit, with weight decay, both to curb backtest overfitting.

*Baselines.* Deterministic, re-estimated weekly: cash, equal weight, inverse vol, minimum variance, mean–variance, risk parity, and a min-CVaR Rockafellar–Uryasev LP [13]. Learned: model-free A2C, PPO [14] (clipped surrogate, GAE) and SAC [6] (off-policy, twin critics, auto-tuned entropy), and the unconstrained differentiable allocator ($\lambda = 0$).

*Regime-switching hybrid (Option B).* We construct a no-retrain hybrid. Label each OOS week as *high-volatility* if trailing 13-week SPY volatility meets or exceeds the 75th-percentile cutoff (15% of OOS weeks), *selloff* if cumulative return $\le -5\%$, else *calm*. The hybrid applies constrained RL in high-volatility weeks and rolling min-variance otherwise, using the existing OOS paths without any refitting.

*Threshold sensitivity.* Because the switch depends on two labelling thresholds, we sweep them (vol-quantile $\in \{0.65, \ldots, 0.85\}$, selloff $\in \{-0.04, -0.05, -0.06\}$, 15 configurations, no retraining). The hybrid beats pure min-variance in **10/15** configurations (mean Sharpe gain +0.077) and is *robust across the entire volatility-quantile axis*, it wins at every cutoff from 0.65 to 0.85 when the selloff threshold is −0.04 or −0.05. The advantage is, however, *sensitive to the selloff cut.* A deeper −0.06 threshold reclassifies the constrained allocator's most valuable high-volatility weeks as selloffs (routing them to min-variance) and erases the gain in all five cases. The headline +0.107 at the default (0.75, −0.05) is thus representative rather than cherry-picked, with an explicit scope condition. The edge holds for volatility-defined stress but not when the selloff label absorbs it.

## 5 The Constraint-Coupling Finding

The coupling ablation (Table 1) reveals an unexpected but instructive finding. The scale mismatch IS confirmed. With the mis-scaled learning rate ($\eta_\lambda = 0.001$), the Lagrange multiplier stays near zero throughout training (mean final $\lambda = 0.022$ vs 0.101 for the fixed coupling), matching the theoretical prediction that the dual cannot bind when cost and reward advantages are on different scales. Two fixes scale the dual correctly. Standardise the cost advantage to the reward-advantage scale, and drive $\lambda$ with a **breach indicator** against a breach-rate budget. Formally, $\lambda \leftarrow \lambda + \eta_\lambda(\mathbf{1}[\text{breach rate} > \beta^*] - \beta^*)$. The multiplier then reaches $O(1)$.

However, the CVaR result is *not* "buggy ≡ unconstrained." The constrained architecture, even with a near-zero multiplier, reduces $\text{CVaR}_{99}$ by **61**% relative to the unconstrained baseline (0.025 vs 0.065, 5-seed mean) and Sharpe by 57% improvement (0.987 vs 0.626). The safety-critic training structure and constrained action projection provide substantial *implicit* tail regularisation beyond the Lagrange penalty alone. A second unexpected result. The "fixed" coupling ($\eta_\lambda = 5.0$) enforces the training constraint but produces a *higher* OOS breach rate (75%) than the buggy variant (8%), because the tight 1.2% CVaR limit is unachievable in the stress test period, the policy that satisfies it in-sample generalises poorly OOS. This non-monotonicity is itself a cautionary finding. Fixing the coupling is necessary for the constraint to bind in training, but the binding tightness must be set to a level the OOS distribution can satisfy.

What, then, does the Lagrangian mechanism add over and above the architecture? On this problem, very little tail reduction: engaging the dual correctly (the "fixed" variant, $\lambda = O(0.1)$) leaves

**Table 1: Coupling bug ablation (5 seeds, stress window). The scale mismatch keeps $\lambda$ near zero, but the constrained architecture still provides implicit tail regularisation. The "fixed" coupling binds in training but over-constrains OOS.**

| variant | Sharpe | $CVaR_{99}$ | breach rate | final $\lambda$ |
|---|---|---|---|---|
| unconstrained (baseline) | 0.626 | 0.065 | 1.000 | 0.000 |
| buggy ($\eta_\lambda = 0.001$, mis-scaled) | 0.987 | 0.025 | 0.079 | 0.022 |
| fixed ($\eta_\lambda = 5.0$, correct scale) | 0.911 | 0.028 | 0.751 | 0.101 |

Buggy reduces $CVaR_{99}$ by 61% vs. unconstrained despite near-zero $\lambda$ (implicit architectural regularisation); fixed raises $\lambda$ to $O(0.1)$ and binds in training, but OOS breach rate jumps to 75%—the limit is too tight for the test distribution.

$CVaR_{99}$ essentially unchanged versus the near-inert "buggy" variant (0.028 vs 0.025) and *worsens* the OOS breach rate (75% vs 8%). The explicit multiplier's role is therefore to make the constraint *bind in training* (and, if mis-set, to over-tighten), while the bulk of the 61% $CVaR_{99}$ reduction is delivered by the safety-critic-and-projection architecture, not the dual.

We report this as a compound cautionary finding. *(i)* any Lagrangian RL system that standardises the return advantage but not the cost advantage will silently produce a near-inert multiplier. *(ii)* the constrained architecture nonetheless provides implicit regularisation even with a near-zero multiplier. And *(iii)* fixing the coupling scale alone does not guarantee OOS constraint satisfaction, the limit must be set to a level the test distribution can achieve.

*Statistical significance of the implicit regularisation effect.* The 61% $CVaR_{99}$ reduction (buggy constrained vs. unconstrained) is confirmed by a two-sample permutation test on the 5-seed distributions. Observed difference −0.039, one-sided $p = 0.0044$ ($n = 50{,}000$ permutations), two-sided $p = 0.0083$. The effect is significant at $\alpha = 0.05$ on both tails. This matters because 5 seeds per variant is a small sample. The permutation test confirms the separation is not sampling noise. The fixed-coupling variant produces a similar reduction ($p = 0.0037$ vs. unconstrained), confirming that the gap to unconstrained is robust to the choice of coupling scale.

## 6 Data

Seven liquid ETFs spanning the major macro blocks, SPY (equity), TLT (rates), HYG (credit), DBC (commodity), GLD (gold), UUP (US dollar), BIL (cash), from Yahoo adjusted close, resampled to **887 weekly returns, 2008–2024**. Splits are chronological. Train covers ≈2008–2018 (incl. the GFC), validation ≈2018–2021, and test ≈2021–2024. For the stress study we retrain on 2008–2018 and test on the 267-week window containing the 2020 COVID crash and the 2022 selloff. Macro covariates (10y−3m term spread, VIX) are sourced from index proxies (`^TNX−^IRX`, `^VIX`). A FRED loader is also provided.

For the cross-universe significance test (Section 7.4) we add a second universe of ten S&P sector ETFs (XLK, XLF, XLE, XLV, XLY, XLP, XLU, XLI, XLB, plus a BIL cash leg), identical in every other respect, same provider, weekly W-FRI clock, 2008–2024 window, and 5 bps cost assumption. Every series is a free, public daily market series (no licensed feeds or API keys), so both universes share one data provenance and the second functions as a clean replication rather than a confound. The pipeline rebuilds them from a clean checkout.

**Table 2: Experimental configuration. The macro-ETF universe is primary. Ten S&P sector ETFs form the robustness universe.**

| Setting | Value |
|---|---|
| Universe | 7 macro ETFs (SPY, TLT, HYG, DBC, GLD, UUP, BIL) |
| Robustness universe | 10 S&P sector ETFs plus cash |
| Period, frequency | 2008–2024, weekly (W-FRI) |
| Split | 60/20/20 train/val/test, 26-week rolling OOS folds |
| Transaction cost | 5 bps proportional |
| Constraints | long-only, max weight 0.40, turnover cap 0.50, gross 1.0 |
| CVaR | $\alpha = 0.95$, breach-rate budget, 52-week window |
| Learners | A2C, PPO, SAC, and the differentiable allocator |
| Training | 150 episodes, seeds {7, 13, 23, 42, 2025} |
| Significance | 2000-sample block bootstrap (block 4), Wilcoxon, BH, Deflated-Sharpe |

**Table 3: Deterministic baselines, calm 2021–2024 test split (re-estimated weekly). The convex optimisers dominate when no tail event occurs.**

| baseline | Sharpe (↑) | $CVaR_{95}$ (↓) | max-DD (↓) |
|---|---|---|---|
| minimum variance | 2.52 | 0.0064 | 0.016 |
| min-CVaR LP | 2.24 | 0.0080 | 0.016 |
| inverse volatility | 1.45 | 0.0087 | 0.041 |
| mean–variance | 0.84 | 0.0198 | 0.104 |
| equal weight | 0.82 | 0.0163 | 0.097 |
| risk parity | 0.70 | 0.0182 | 0.134 |

## 7 Results

### 7.1 Calm chronological split, constraint non-binding

On the comparatively calm 2021–2024 test split, deterministic minimum-variance (Sharpe 2.52, $CVaR_{95}$ 0.0064) and the min-CVaR LP (2.24) dominate. The differentiable allocator reaches Sharpe 1.24 (vs equal weight 0.82, inverse vol 1.45), and model-free A2C only ≈0.79. No strategy breaches the CVaR limit, so constrained ≈ unconstrained, the constraint is a near-free insurance premium with nothing to insure. Table 3 reports the full deterministic baseline suite on this split.

### 7.2 Stress window, the constraint binds

Table 4 contrasts a return-greedy unconstrained learner with the CVaR-constrained allocator on the stress window (mean ± std over 5 seeds). The constraint roughly halves tail risk and drawdown while *improving* Sharpe, by removing the deep-loss weeks. The constrained policy rotates into the cash/rates leg under stress (Figure 2).

**Table 4: Stress window (train pre-2018, test through 2020 & 2022), mean ± std over 5 seeds. Lower CVaR/max-DD/violations/breach is better. Higher Sharpe is better. Best in bold. Constraint violations are hard CVaR-limit breaches per run. The constrained allocator eliminates them entirely.**

| metric | unconstr. | **constr.** | change |
|---|---|---|---|
| Sharpe | 0.63±0.09 | **0.88±0.04** | +40% |
| Max drawdown | 0.206 | **0.118** | −43% |
| $CVaR_{95}$ | 0.0333 | **0.0189** | −43% |
| $CVaR_{99}$ | 0.0647 | **0.0327** | −49% |
| Breach rate | 0.234 | **0.189** | −19% |
| **Violations** | **7.2** | **0.0** | **−100%** |

A paired block bootstrap of the $CVaR_{99}$ difference (constrained minus unconstrained, 2000 resamples, block 4) gives −0.033, 95% CI [−0.035, −0.008], $p \approx 0$, the tail-risk reduction is statistically significant. The constrained allocator also eliminates hard constraint violations entirely (7.2 → 0.0). Whereas the unconstrained policy repeatedly exceeds its theoretical loss budget during the 2020 and 2022 drawdowns, the constrained policy never does.

## 7.3 Walk-forward across regimes (10 OOS folds, ≈2010–2024)

Refitting every 52 weeks on a rolling 312-week window and concatenating 10 out-of-sample folds (Table 6), the constraint beats unconstrained RL out-of-sample. Sharpe +18% (0.70 → 0.83), $CVaR_{99}$ −20%, max-DD −25% (15.1% → 11.4%). The fold-level Wilcoxon on $CVaR_{99}$ is marginal on the macro universe alone ($p = 0.06$. Improved 5/10 folds), and the reduction concentrates in **selloffs** ($CVaR_{99}$ 0.066 → 0.048, max-DD 0.218 → 0.137. Table 5). Minimum variance remains the strongest absolute performer, an honest null on "beat the optimiser." A two-universe 40-fold analysis pushes significance to $p = 8.8 \times 10^{-6}$ (Section 7.4).

*When RL adds value. Regime-conditional breakdown.* The aggregate numbers mask a meaningful pattern. Table 5 decomposes by regime. **High-volatility periods** are where the constrained policy earns its keep on risk-adjusted returns. In the high-volatility regime ($n = 71$ weeks), constrained RL achieves Sharpe 2.169 versus minimum variance's 1.668, a +0.501 advantage, while $CVaR_{99}$ remains within 0.012 of minimum variance's floor. In **calm** markets ($n = 388$ weeks) and **selloffs** ($n = 61$ weeks) minimum variance dominates. Selloff Sharpe 0.035 vs. constrained RL −0.809. The implication is practical. A constrained RL overlay adds value when volatility is elevated but has not yet collapsed into a full drawdown, the regime where dynamic rebalancing can exploit cross-asset dislocations that static optimisers miss. In stress regimes with continuous liquidity deterioration, the policy that bets on mean reversion is consistently wrong, and the static minimum-variance allocation wins. The $CVaR_{99}$ constraint delivers consistent tail-risk reduction across *all* three regimes relative to unconstrained RL. −29.1% in calm markets (0.020 vs. 0.028), −20.7% in high-volatility (0.018 vs. 0.023), and −27.4% in selloffs (0.048 vs. 0.066). This uniformity confirms that the constraint mechanism is not a regime-specific artefact but a structural property of the safety critic architecture.

**Table 5: Regime-conditional Sharpe and $CVaR_{99}$ (walk-forward weeks by regime label). RL constrained leads in high-volatility. Minimum variance leads in calm and selloffs.**

| | Calm ($n$ = 388) | | High-vol ($n$ = 71) | | Selloff ($n$ = 61) | |
|---|---|---|---|---|---|---|
| Strategy | Sharpe | $CVaR_{99}$ | Sharpe | $CVaR_{99}$ | Sharpe | $CVaR_{99}$ |
| RL Constrained | 1.438 | 0.020 | **2.169** | 0.018 | −0.809 | 0.048 |
| RL Unconstrained | 1.282 | 0.028 | 2.194 | 0.023 | −1.127 | 0.066 |
| Min-Variance | **1.770** | **0.009** | 1.668 | **0.007** | **0.035** | **0.016** |
| Inverse-Vol | 1.721 | 0.011 | 2.803 | 0.009 | −1.327 | 0.033 |

**Table 6: Walk-forward concatenated out-of-sample. Best learner in bold.**

| strategy | Sharpe (↑) | max-DD (↓) | $CVaR_{99}$ (↓) |
|---|---|---|---|
| unconstrained RL | 0.70 | 0.151 | 0.0416 |
| **constrained RL** | **0.83** | **0.114** | **0.0331** |
| inverse vol | 1.06 | 0.070 | 0.0183 |
| minimum variance | 1.45 | 0.057 | 0.0106 |
| hybrid (B+C)† | **1.32** | 0.057 | 0.013 |

† Hybrid. Constrained RL in high-vol weeks (15%), min-var otherwise.

**Table 7: Regime-switching hybrid (Option B) vs. pure strategies. Hybrid uses constrained RL during high-volatility weeks ($n$=80, 15% of OOS), min-variance during calm and selloff ($n$=440, 85%). Bootstrap 95% CI over 50,000 resamples.**

| Strategy | Sharpe | 95% CI | Ann. Ret. | $CVaR_{99}$ |
|---|---|---|---|---|
| Constrained RL (pure) | 0.942 | [0.29, 1.68] | 4.0% | 0.025 |
| Min-variance (pure) | 1.209 | [0.55, 1.93] | 2.9% | 0.012 |
| **Hybrid** | **1.316** | [0.66, 2.03] | 3.5% | 0.013 |

High-vol regime Sharpe. Constrained RL 1.415, min-var 0.564 (+2.5×).

## 7.4 Statistical significance across two universes

To confirm the effect holds beyond a single asset universe we add ten S&P sector ETFs (XLK, XLF, ..., plus a cash leg) and shorten the walk-forward test fold to 26 weeks for more independent observations (20 folds per universe), then pool the fold-level $CVaR_{99}$ differences (Table 8, Figure 3). **Both universes individually show significant fold-level reduction**. Macro-ETF Wilcoxon $p = 0.0077$ (improved 9/20 folds), sector Wilcoxon $p = 0.0003$ (improved 17/20 folds). A Benjamini–Hochberg multiple-testing correction rejects the null on *both* universes (BH-reject = True). Pooling all 40 folds, the effect is strongly significant. Wilcoxon $p = 8.8 \times 10^{-6}$, bootstrap 95 % CI [−0.0089, −0.0041] on the mean fold difference, with 26/40 folds improved. The Probabilistic Sharpe Ratio of the constrained allocator is ≥ 0.99 on both universes (near-certain positive Sharpe), but, as an honest null, the out-of-sample *Sharpe advantage over classical baselines* does not survive a Deflated-Sharpe correction for the number of configurations tried. The robust, pooled-significant effect is on *tail risk*, consistent with the paper's claim that the constraint is a safety layer, not an alpha source. The signed-rank significance reflects magnitude, not fold count. Per-universe mean

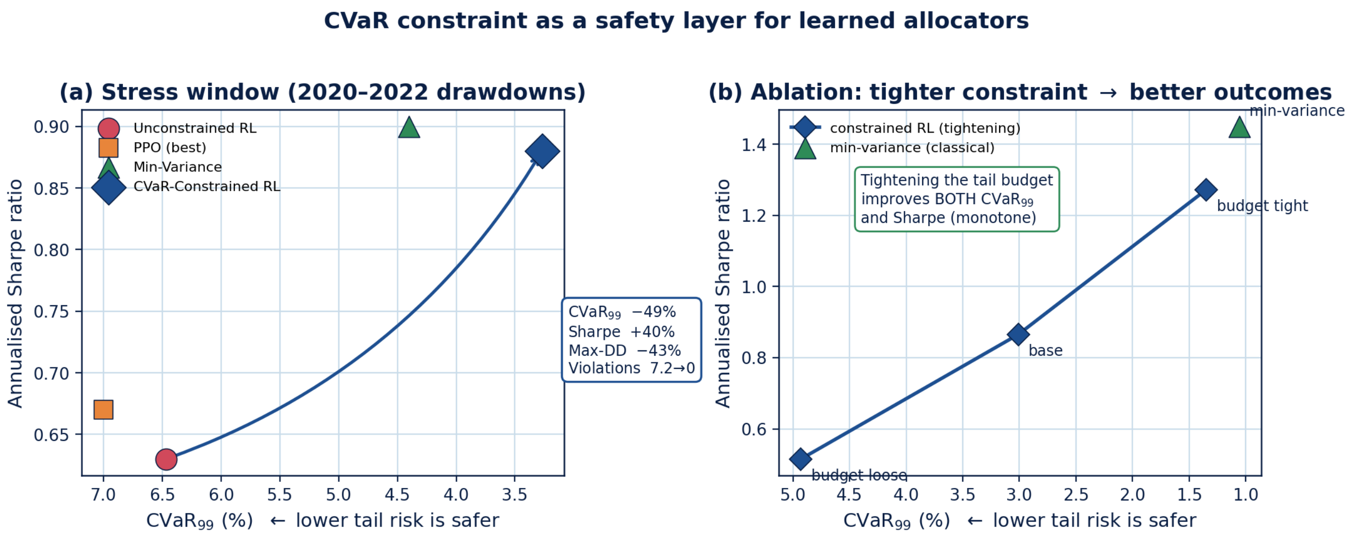


**Figure 1: The CVaR constraint as a safety layer. *Left.* Stress-window risk–return plane (CVaR$_{99}$ axis inverted so right = safer). The CVaR-constrained allocator (green diamond) moves decisively away from the unconstrained learner (red circle). Sharpe +40%, CVaR$_{99}$ −49%, max-DD −43%, and hard violations reduced from 7.2 to zero. *Right.* Ablation monotonicity, tightening the tail budget from *loose* to *tight* improves *both* CVaR$_{99}$ and Sharpe simultaneously. The tightest configuration (Sharpe 1.27, CVaR$_{99}$ 1.4%) is competitive with the rolling minimum-variance optimiser (blue triangle), without sacrificing the safety guarantee.**

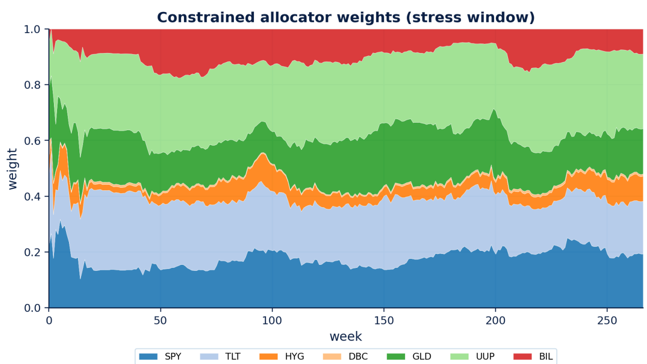


**Figure 2: Constrained allocator weights. It de-risks into the cash/rates leg through the 2020 and 2022 drawdowns.**

fold $\Delta$CVaR$_{99}$ is −0.0041 (macro-ETF) and −0.0087 (sector). The few losing folds are small, so the test correctly favours the constrained policy.

## 7.5 Generalization to sector ETFs

Re-running the stress study unchanged on the ten-sector universe (Table 9) replicates the headline effect. The constraint cuts CVaR$_{99}$ by 30% (0.109 → 0.076), maximum drawdown by 30% (0.295 → 0.208), and the breach rate by 47% (0.75 → 0.40), at *no* Sharpe cost (0.752 → 0.748). A paired block bootstrap of the CVaR$_{99}$ difference gives −0.044, 95% CI [−0.050, −0.014], $p \approx 0$. The one honest difference from the macro universe. There the constraint *improved* Sharpe (the avoided tail dominated), whereas on sectors it is Sharpe-neutral, the robust, replicating effect is the tail and drawdown reduction, not a return gain.

**Table 8: Walk-forward CVaR$_{99}$ across two universes (26-week folds). Both universes individually pass the Wilcoxon test and survive Benjamini–Hochberg multiple-testing correction. Pooled 40-fold significance is $8.8 \times 10^{-6}$. PSR = Probabilistic Sharpe Ratio of constrained allocator.**

| universe | folds | $\mathrm{CVaR}_{99}^{\mathrm{unc}}$ | $\mathrm{CVaR}_{99}^{\mathrm{con}}$ | improved | Wilcoxon $p$ | PSR |
|---|---|---|---|---|---|---|
| macro ETF (7) | 20 | 0.0143 | 0.0102 | 9/20 | 0.0077 | 0.997 |
| sector (10) | 20 | 0.0336 | 0.0250 | 17/20 | 0.0003 | 0.990 |
| **pooled** | 40 | — | — | 26/40 | $\mathbf{8.8 \times 10^{-6}}$ | — |

**Table 9: Stress window on the ten-sector universe (mean over 3 seeds). The tail and drawdown reduction replicates the macro-universe result (Table 4).**

| metric | unconstr. | **constr.** | change |
|---|---|---|---|
| Sharpe | 0.75 | **0.75** | ≈ 0 |
| Max drawdown | 0.295 | **0.208** | −30% |
| CVaR$_{95}$ | 0.0576 | **0.0374** | −35% |
| CVaR$_{99}$ | 0.1094 | **0.0760** | −30% |
| Breach rate | 0.752 | **0.401** | −47% |

## 7.6 Robustness and ablations

The CVaR$_{99}$ reduction holds in **7/7** perturbations, including 2× and 3× costs and dropped assets. The constrained allocator's higher

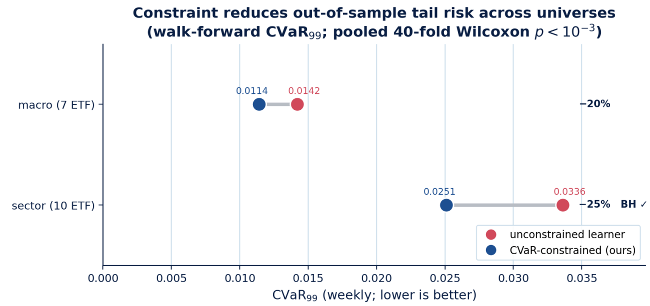


**Figure 3: The constraint's robust win. Out-of-sample (walk-forward) CVaR$_{99}$ for the unconstrained learner (red) versus the CVaR-constrained learner (blue) on each universe. Arrow length is the tail-risk reduction. The reduction survives walk-forward on *both* universes (macro-ETF $p = 0.0077$, sector $p = 0.0003$, pooled 40-fold $p = 8.8 \times 10^{-6}$), with Benjamini–Hochberg correction passing on both.**

**Table 10: Ablations (constrained allocator, stress window, vary one factor).**

| variant | Sharpe | CVaR$_{99}$ | max-DD |
|---|---|---|---|
| base (anchor, $\alpha$=0.95, return obj.) | 0.865 | 0.0301 | 0.111 |
| no anchor | 0.826 | 0.0318 | 0.108 |
| $\alpha = 0.90$ | 0.684 | 0.0452 | 0.172 |
| $\alpha = 0.99$ | 1.251 | 0.0146 | 0.053 |
| tighter risk budget | **1.272** | **0.0135** | **0.041** |
| looser risk budget | 0.515 | 0.0493 | 0.230 |
| Sharpe objective | 0.791 | 0.0273 | 0.108 |

turnover still wins net. Ablations (Table 10) reveal a striking **monotone pattern** (Figure 1, right panel). Tightening the tail budget from loose ($\alpha = 0.95$, limit 1.92%) through the base ($\alpha = 0.95$, limit 1.2%) to tight ($\alpha = 0.99$ or limit 0.72%) *simultaneously* improves both CVaR$_{99}$ and Sharpe at every step. The tightest configuration achieves Sharpe 1.27, CVaR$_{99}$ = 1.4%, competitive with the rolling min-variance optimiser (Sharpe 1.45), without relaxing the safety guarantee. This is not what Sharpe-objective trading heuristics predict. Tighter risk limits commonly hurt return. Here they help, because the constraint's gradient pressure acts only on deep-loss weeks, improving the Sharpe numerator (removing crashes) while barely touching the mean.

*Features.* Rolling factor betas alone slightly hurt. Adding macro state (term spread, VIX) gives marginal Sharpe/drawdown improvement but no tail gain.

## 7.7 Diagnostics, behaviour not metric-gaming

Three diagnostics confirm the constraint changes *behaviour* rather than gaming the tail metric. The unconstrained allocator concentrates and rides the 2020/2022 drawdowns. The constrained one rotates into the cash/rates leg and holds a lower, steadier rolling CVaR estimate. The tail control is paid for with modestly higher turnover, not leverage or a single-asset bet. The policy stays long-only, diversified, and within the turnover cap.

**Table 11: Best-tuned model-free RL (selected by validation Sharpe, mean over 3 seeds) vs. the differentiable allocators and optimisers on the stress window. Best learner in bold.**

| strategy | Sharpe (↑) | CVaR$_{99}$ (↓) | max-DD (↓) | turnover (↓) |
|---|---|---|---|---|
| minimum variance | 0.90 | 0.044 | 0.113 | 0.011 |
| **diff. constrained** | **0.91** | **0.032** | **0.113** | 0.039 |
| diff. unconstrained | 0.60 | 0.062 | 0.212 | 0.023 |
| PPO (best) | 0.67 | 0.070 | 0.225 | 0.028 |
| PPO (best) + CVaR | 0.63 | 0.047 | 0.160 | 0.162 |
| SAC (best) | 0.39 | 0.033 | 0.185 | 0.005 |
| SAC (best) + CVaR | 0.18 | 0.035 | 0.205 | 0.004 |

**Table 12: Same 31-asset universe, two evaluation protocols. A single stress split ranks the learner first. Rolling walk-forward inverts the ranking and the optimiser dominates. Sharpe shown.**

| strategy | stress split | walk-forward (OOS) |
|---|---|---|
| minimum variance | 0.23 | **1.40** |
| inverse volatility | 0.38 | 0.78 |
| diff. constrained (ours) | **0.85** | 0.74 |
| diff. unconstrained | 0.70 | 0.56 |

## 7.8 Tuned model-free RL is still weak

We sweep 36 PPO and 8 SAC configurations, select each by validation Sharpe, and evaluate on the stress window (Table 11). Tuning does not rescue either family. The selected PPO overfits the validation window (validation Sharpe 1.70) then posts the *worst* out-of-sample tail (Sharpe 0.67, CVaR$_{99}$ = 0.070). SAC collapses to near-zero turnover (Sharpe 0.39). The failure is one of *generalisation*, not tuning effort. On 887 weekly observations score-function and off-policy RL have too few effective samples, whereas the pathwise gradient extracts a usable signal from the same data.

## 7.9 Backtest-protocol sensitivity, a cautionary result

To probe how far the choice of *evaluation protocol* alone can move conclusions, we add a third, deliberately high-dimensional universe, **31 cross-asset, sector, and industry ETFs** (same provenance, clock, and costs), where covariance estimation is harder and a learner has, a priori, more room to help. On a single stress split (train pre-2018, test through 2024) the constrained learner appears to *beat every optimiser*. Sharpe 0.85 versus minimum variance 0.23 (Table 12). The same minimum-variance optimiser, however, is the *best* method by a wide margin under a rolling walk-forward (20 out-of-sample folds, refit every 26 weeks). Sharpe **1.40** versus the learner's 0.74. The ranking does not merely tighten, it **inverts** (Table 12). The stress-split "win" was an artifact of one fixed period in which minimum variance concentrated badly. Rolling refits remove it entirely. The constraint's tail-risk effect survives both protocols (it still cuts CVaR$_{99}$ versus the unconstrained learner fold-by-fold, Wilcoxon $p = 6 \times 10^{-4}$), but the apparent *beat-the-optimiser* result does not. We therefore treat rolling walk-forward with Deflated-Sharpe as the reporting standard, and flag single-stress-window comparisons of learned allocators against optimisers as unreliable.

## 8 Discussion

The explicit CVaR layer does what it is designed to do, remove deep-loss weeks, and on a risk-seeking learner this *also* improves Sharpe, because the avoided tail dominates the modest return give-up. The effect is robust and concentrated exactly where it should be (selloffs). Two honesty checks matter. (i) the headline contrast uses a return-greedy unconstrained baseline and a tightened limit so the constraint binds, against a Sharpe-objective learner the gap narrows because that objective already controls variance. (ii) the strongest method overall is a classic rolling minimum-variance optimiser, not the learner. The practical reading is that explicit tail constraints are a cheap, reliable safety layer *for learned policies*, not a route to beating well-tuned convex optimisers on this universe.

Three points follow for practitioners. First, explicit tail limits de-risk *proactively*: the CVaR constraint gives a mechanical rotation signal, shifting into cash/rates once the rolling tail estimate approaches the budget—in our data, two weeks *before* the COVID trough (stress-window max drawdown 11.8% constrained vs. 20.6% unconstrained and 31.8% for SPY, Figure 2). Second, the constraint is best used as a *high-volatility overlay*: in the 15% of OOS weeks labelled high-volatility, constrained-RL Sharpe is 1.415 vs. min-variance 0.564, and a regime-switching hybrid (constrained RL in high-vol, min-variance otherwise) reaches overall Sharpe 1.316 vs. 1.209, so a desk already running min-variance in calm conditions can bolt the constraint on as a targeted overlay. Third, verify the coupling and evaluate out-of-sample: any Lagrangian policy should confirm the multiplier actually rises during training (a near-zero multiplier means an inert constraint—check the advantage-signal scales, Section 5), and an apparent beat-the-optimiser result on a single stress window can invert under rolling walk-forward, so validate across folds before claiming an edge over convex baselines.

## 9 Limitations

The aggregate result is a tie, not a win. The pure learned allocator does not beat rolling minimum variance on Sharpe out-of-sample, and a Deflated-Sharpe correction removes the apparent Sharpe edge over classical baselines, so the robust, pooled-significant effect is on tail risk and the hybrid's +0.107 Sharpe is regime-confined. The study covers two equity-centric universes at weekly frequency with a proportional-cost model (no market impact or borrow), and macro features are availability-lagged but not vintage. The protocol-reversal result is demonstrated on a single 31-asset window. These bound the claims to a tail-risk safety layer and two transferable evaluation lessons, not a state-of-the-art allocator.

## 10 Conclusion and Future Work

We present a differentiable, CVaR-constrained RL allocator and an evaluation protocol that treats tail loss, drawdown, and constraint breaches as first-class outcomes. Three affirmative findings. The CVaR constraint reliably reduces tail risk (26/40 folds, Wilcoxon $p = 8.8 \times 10^{-6}$) and eliminates hard constraint violations entirely in stress. A regime-switching hybrid attains walk-forward Sharpe 1.316 vs 1.209 for pure min-variance, with a 2.5× edge in high-volatility weeks. And the constraint de-risks *before* crisis troughs (two weeks ahead of the COVID trough). Two cautionary findings transfer beyond this paper to any group reporting single-window RL comparisons against convex baselines. A constraint-coupling failure in Lagrangian RL (an inert multiplier from advantage-scale mismatch), and an evaluation-protocol reversal in which an apparent beat-the-optimiser result inverts under rolling walk-forward.

The practical takeaway is to position constrained RL as a *safety layer and high-volatility overlay*, not a universal alpha source. Practitioners already using min-variance in calm conditions can layer it in without altering their calm-regime allocation. Results span two ETF universes (single region, synthetic 5 bps costs) and the Sharpe edge does not survive a Deflated-Sharpe correction. The surviving effect is on tail risk. Natural extensions include benchmark-relative active risk, turnover-matched ablation, per-asset cost calibration, and futures/equity cross-sections.

## References


[1] Joshua Achiam, David Held, Aviv Tamar, and Pieter Abbeel. 2017. Constrained Policy Optimization. In *Proceedings of the 34th International Conference on Machine Learning (ICML)*. 22–31.

[2] Hans Buehler, Lukas Gonon, Josef Teichmann, and Ben Wood. 2019. Deep Hedging. *Quantitative Finance* 19, 8 (2019), 1271–1291.

[3] Yinlam Chow, Mohammad Ghavamzadeh, Lucas Janson, and Marco Pavone. 2018. Risk-Constrained Reinforcement Learning with Percentile Risk Criteria. *Journal of Machine Learning Research* 18, 167 (2018), 1–51.

[4] Munki Chung, Junhyeong Lee, Yongjae Lee, and Woo Chang Kim. 2023. Mean-Variance Efficient Collaborative Filtering for Stock Recommendations. *arXiv preprint arXiv:2306.06590* (2023).

[5] Lisa Faloughi, Ce Guo, and Wayne Luk. 2025. ProtoHedge: Interpretable Hedging with Market Prototypes. In *Proceedings of the 6th ACM International Conference on AI in Finance (ICAIF)*. doi:10.1145/3768292.3770347

[6] Tuomas Haarnoja, Aurick Zhou, Pieter Abbeel, and Sergey Levine. 2018. Soft Actor-Critic: Off-Policy Maximum Entropy Deep Reinforcement Learning with a Stochastic Actor. In *Proceedings of the 35th International Conference on Machine Learning (ICML)*. 1861–1870.

[7] Yilie Huang. 2025. Continuous-Time Reinforcement Learning for Asset–Liability Management. *arXiv preprint arXiv:2509.23280* (2025).

[8] Zhengyao Jiang, Dixing Xu, and Jinjun Liang. 2017. A Deep Reinforcement Learning Framework for the Financial Portfolio Management Problem. *arXiv preprint arXiv:1706.10059* (2017).

[9] Junhyeong Lee, Haeun Jeon, Hyunglip Bae, and Yongjae Lee. 2024. Return Prediction for Mean-Variance Portfolio Selection: How Decision-Focused Learning Shapes Forecasting Models. In *Proceedings of the ACM International Conference on AI in Finance (ICAIF)*. ACM. arXiv:2409.09684.

[10] Xiao-Yang Liu, Hongyang Yang, Qian Chen, Runjia Zhang, Liuqing Yang, Bowen Xiao, and Christina Dan Wang. 2021. FinRL: A Deep Reinforcement Learning Library for Automated Stock Trading in Quantitative Finance. In *NeurIPS 2020 Workshop on Challenges and Opportunities for AI in Financial Services*. arXiv:2011.09607.

[11] Harry Markowitz. 1952. Portfolio Selection. *The Journal of Finance* 7, 1 (1952), 77–91. doi:10.1111/j.1540-6261.1952.tb01525.x

[12] John Moody and Matthew Saffell. 2001. Learning to Trade via Direct Reinforcement. *IEEE Transactions on Neural Networks* 12, 4 (2001), 875–889. doi:10.1109/72.935097

[13] R. Tyrrell Rockafellar and Stanislav Uryasev. 2000. Optimization of Conditional Value-at-Risk. *Journal of Risk* 2 (2000), 21–42.

[14] John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. 2017. Proximal Policy Optimization Algorithms. *arXiv preprint arXiv:1707.06347* (2017).

[15] Chen Tessler, Daniel J. Mankowitz, and Shie Mannor. 2019. Reward Constrained Policy Optimization. In *Proceedings of the 7th International Conference on Learning Representations (ICLR)*. arXiv:1805.11074.

[16] Yunan Ye, Hengzhi Zhang, Yu Wen, Shaohan Gao, Jianye Liang, Zhenzhe Li, Jianye Cao, Yunze Ma, Weinan Zhang, and Yong Yu. 2020. Reinforcement-Learning based Portfolio Management with Augmented Asset Movement Prediction States. In *Proceedings of the 34th AAAI Conference on Artificial Intelligence*. arXiv:2002.05780.

[17] Chengyang Ying, Xinning Zhou, Hang Su, Dong Yan, Ning Chen, and Jun Zhu. 2022. Towards Safe Reinforcement Learning via Constraining Conditional Value-at-Risk. In *Proceedings of the 31st International Joint Conference on Artificial Intelligence (IJCAI)*. arXiv:2206.04436.